An injective C*-power series violating the Bieberbach coefficient bound

Research proof and Lean formalization

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Abstract

We give a normalized power series on $C([0,1],\mathbb{C})$ which converges in norm at every element and is globally injective, but whose second coefficient violates the proposed C*-algebraic Bieberbach order bound. Its formal derivative series also converges everywhere. Injectivity follows from a continuous exponential period anchored at zero; injectivity of individual scalar fibers is not assumed.

Krishna's Conjecture 1.2 in [1] proposes the following extension of the scalar Bieberbach theorem. If $\mathcal{A}$ is a unital C*-algebra and $F(u) = u + \sum_{n\geq 2} a_n u^n$ converges in norm and is injective on $\{u : \|u\| < 1\}$, then

$$a_n a_n^* \leq n^2 1, \qquad a_n^* a_n \leq n^2 1 \qquad (n \geq 2). \tag{1}$$

Conjecture 1.4 adds norm convergence of the derivative series. The next example violates both general-coefficient assertions. The separate invertible-coefficient variants are not claimed refuted.

Theorem 1. *Let $\mathcal{A} = C([0,1],\mathbb{C})$ with its supremum norm, and let $a(t) = 8t$. Put*

$$c_n = \frac{a^{n-1}}{n!} \quad (n \geq 1), \qquad F(u) = \sum_{n\geq 1} c_n u^n. \tag{2}$$

This series and its formal derivative series converge in the norm of $\mathcal{A}$ for every $u \in \mathcal{A}$. The map $F : \mathcal{A} \to \mathcal{A}$ is injective. Nevertheless $c_2(1) = 4$, so both inequalities in (1) fail at $n = 2$.

Proof. For every $u \in \mathcal{A}$,

$$\sum_{n\geq 1} \|c_n u^n\| \leq \|u\| \sum_{k\geq 0} \frac{\|au\|^k}{(k+1)!} < \infty.$$

The derivative series is dominated by an exponential series as well. Multiplication of (2) by a and reindexing the absolutely convergent exponential series gives the identity in $\mathcal{A}$

$$aF(u) = \exp(au) - 1. \tag{3}$$

Here the exponential is the Banach-algebra exponential, whose scalar evaluation is the usual complex exponential. Evaluation of the norm-convergent series at $t = 0$ also gives $F(u)(0) = u(0)$. In particular,

$$F(u)(t) = \frac{\exp(8tu(t)) - 1}{8t} \quad (t > 0).$$

Suppose $F(u) = F(v)$. By (3), the continuous function $h(t) = 8t(u(t) - v(t))$ satisfies $\exp(h(t)) = 1$ for every t, and $h(0) = 0$. Thus its image lies in the discrete set $2\pi i\mathbb{Z}$. Connectedness of $[0,1]$

forces $h = 0$. Since $8t \neq 0$ for $t > 0$, we obtain $u(t) = v(t)$ there, and the equality at 0 follows from $F(u)(0) = F(v)(0)$. Hence $u = v$.

Finally $c_1 = 1$ and $c_2(t) = 4t$. Evaluation of either claimed order bound at $t = 1$ would give $16 \leq 4$, a contradiction. □

Why the scalar theorem does not apply. The source requires injectivity of a map on the unit ball of the algebra. It does not require that every scalar fiber be univalent. A collision of two continuous inputs in this example would produce a continuous integer-valued exponential period, which must vanish because it vanishes at the endpoint. Some individual scalar fibers are noninjective. The classical scalar Bieberbach theorem is therefore not contradicted. The coefficients c_n for $n \geq 2$ vanish at 0 and are not invertible elements of $\mathcal{A}$; the source's invertible-coefficient restrictions cannot be silently discarded.

Formal scope and review status. The Lean certificate has 28 declarations in four modules. It proves actual Banach-norm summability, the exponential identity, scalar evaluation, global injectivity, the literal coefficient-order violation, and closed negations of the two source assertions specialized to the genuine commutative unital C*-algebra $C([0, 1], \mathbb{C})$. No convergence, injectivity or fiber-choice hypothesis remains. The endpoints are

`source_counterexample`, `source_strong_counterexample`

in namespace `CStarBieberbach`. The isolated strict replay uses Lean 4.34.1 and mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`,

with only `propext`, `Classical.choice` and `Quot.sound`. It checks the fixed coefficient $8t$; no separately formalized theorem about all parameter values is claimed.

Independent preliminary source, mathematical, bounded prior-art and substantive-eligibility review passes; complete final review is pending. The source was read in the author-deposited Zenodo version (doi:10.5281/zenodo.6917925), whose text agrees with the source reading of Preprints version 1. Direct Preprints PDF retrieval remains unavailable. Discovery credit remains zero. The two variants use one construction and constitute at most one prospective outcome. Classical exponential-series and connectedness facts and helper declarations are not additional discoveries. No worldwide priority claim is made.

References

[1] K. Mahesh Krishna, *C*-algebraic Bieberbach, Robertson, Lebedev–Milin, Zalcman, Krzyz and Corona Conjectures*, Preprints (2022), version 1, Conjectures 1.2 and 1.4, pp. 2–3. doi:10.20944/preprints202208.0140.v1.